The cutting plane algorithm with Gomory fractional cuts

Outline

- Why a cutting plane algorithm?
- Gomory fractional cuts
  - Validity of the Gomory fractional cuts
  - Inclusion of Gomory fractional cuts into the simplex dictionary
- A cutting plane algorithm based on Gomory fractional cuts
  - Cut generation operations
  - High-level pseudocode
  - Re-optimization with the dual simplex algorithm
  - Cut selection strategies
  - Heuristic initialization
  - Additional remarks
- The car example
- Typical mistakes
- In summary

Learning objectives

At the end of this chapter you will be able to:

- generate a Gomory fractional cut from a row of a simplex dictionary, and prove that it is valid and that it cuts off the current basic solution;
- insert a cut into the dictionary and re-optimize with the dual simplex algorithm;
- execute a cutting plane algorithm by hand, with the largest right-hand side selection strategy and its tie-breaking rules;
- compute the upper bound at every iteration, and explain why the algorithm may need many cuts;
- explain why these cuts are valid only for ILP problems with integer data.

Prerequisites

- From the module *Linear Programming*, Chapter 10 «Dual Simplex algorithm»: simplex dictionaries, dual feasible bases and dual pivoting operations. A short recap is given in this chapter
- From this module, the chapters «Integer linear programming» and «Primal and dual bounds»: ILP problems, the LP relaxation, and the upper bound given by the LP relaxation
- From this module, the chapter «Valid inequalities»: valid inequalities and cuts that remove fractional points

Why a cutting plane algorithm?

Introduction: Why a cutting plane algorithm?

- The LP relaxation gives an upper bound, but its optimal solution is often fractional
- We go back to the car example and look at its fractional LP solution
- The idea of this chapter: cut off the fractional point, keep all the integer points, and solve again

Back to the car example

- We start from the **car example**:

$$\begin{array}{rrcrcr} \max & 3\,x_1 & + & 2\,x_2 & & \\ \text{subject to} & 10\,x_1 & + & 4\,x_2 & \leq & 45 \\ & 4\,x_1 & + & 14\,x_2 & \leq & 49 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

- The optimal solution of its LP relaxation is the **fractional** point $(\frac{7}{2}, \frac{5}{2})$, with value $\frac{31}{2}$

The idea

Add inequalities that **cut off** this point
without removing any integer feasible solution,

and solve the LP relaxation **again**.

- Repeating these two steps is a **cutting plane algorithm**
- The inequalities of this chapter are the **Gomory fractional cuts**, read directly from the simplex dictionary

A challenge

Before going on, try to guess
how many cuts are needed to reach
the optimal integer solution of the car example.

The last sections answer the question.

Summary

- The LP relaxation of the car example has the fractional optimal solution $(\frac{7}{2}, \frac{5}{2})$, of value $\frac{31}{2}$
- A **cut** removes such a point and keeps all the integer feasible solutions
- Cut, solve again, and repeat: this is the cutting plane algorithm

Gomory fractional cuts

Introduction: Gomory fractional cuts

- We define cutting planes and the Gomory fractional cuts, generated from a row of a simplex dictionary
- We prove that every such cut removes the current fractional basic solution and no integer feasible solution
- We see how to write a cut in the original variables, make it primitive and insert it into the dictionary
- These are the building blocks of the cutting plane algorithm

Definition 1: cutting plane

Given an ILP problem and a fractional optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP})$ *of its LP relaxation, a* ***cutting plane*** *(or* ***cut****) is a valid inequality for the ILP problem that cuts off* $\tilde{\boldsymbol{x}}(\mathrm{LP})$*.*

Gomory fractional cuts

Gomory fractional cuts are valid inequalities
that cut off fractional basic feasible solutions
of the LP relaxation of an ILP problem
without excluding any integer feasible solution.

Integer input data

- **We consider ILP problems in canonical maximization form with integer input data**
- For these problems, the slack variables introduced in the standard-form LP relaxation also take integer values in every integer feasible solution of the ILP problem
- Gomory fractional cuts are valid only if all the variables, slack variables included, are integer
- For MIP problems with continuous variables they can cut off feasible solutions, and a different family of cuts is needed

Recall: simplex dictionaries

- A simplex dictionary of the LP relaxation in standard form reads as follows

$$\begin{cases} x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij} x_j, & \forall i \in \mathcal{B}, \\ z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j x_j, \end{cases}$$

  where $\mathcal{B}$ is the set of indices of the basic variables and $\mathcal{N}$ is the set of indices of the non-basic variables

Recall: basic feasible solutions

- If $\bar{b}_i \geq 0$ for every $i \in \mathcal{B}$, then the dictionary is associated with the following basic feasible solution:

$$x_i = \bar{b}_i, \quad \forall i \in \mathcal{B}, \qquad x_j = 0, \quad \forall j \in \mathcal{N}.$$

- If, in addition, at least one $\bar{b}_i$ is fractional, then this basic feasible solution is **fractional**

- Gomory fractional cuts are obtained from the rows of a simplex dictionary associated with a fractional basic feasible solution of the LP relaxation in standard form of an ILP problem

Definition 2: Gomory fractional cut

Given a row $i \in \mathcal{B}$ of a simplex dictionary associated with a basic feasible solution such that $\bar{b}_i$ is fractional, the linear inequality

$$\sum_{j \in \mathcal{N}} (\bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor) x_j \geq \bar{b}_i - \lfloor \bar{b}_i \rfloor$$

*is called the **Gomory fractional cut** generated by row i.*

- We will prove that any Gomory fractional cut does not cut off any integer feasible solution of the ILP problem and that it cuts off the fractional basic feasible solution

Before the proofs, we see the cuts at work on a small ILP problem with two variables.

Example (1/4)

- Consider the following ILP problem in canonical maximization form with 2 integer variables and 2 linear constraints (called Example 1):

$$\begin{array}{rrcrcr} \max & 2\,x_1 & + & 4\,x_2 & & \\ \text{subject to} & -\,x_1 & + & x_2 & \leq & 2 \\ & 8\,x_1 & + & 2\,x_2 & \leq & 19 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

Example (2/4)

- The LP relaxation in standard form with 4 non-negative variables (2 slack variables) and 2 linear constraints reads as follows:

$$
\begin{array}{rrcrcrcrcr}
\max & 2\,x_1 & + & 4\,x_2 & & & & & & \\
\text{subject to} & -\,x_1 & + & x_2 & + & x_3 & & & = & 2 \\
& 8\,x_1 & + & 2\,x_2 & & & + & x_4 & = & 19 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

Example (3/4)

- Consider the following simplex dictionary associated with a fractional basic feasible solution of the LP relaxation in standard form:

$$\begin{cases} x_1 = \frac{3}{2} + \frac{1}{5} x_3 - \frac{1}{10} x_4 \\ x_2 = \frac{7}{2} - \frac{4}{5} x_3 - \frac{1}{10} x_4 \\ z = 17 - \frac{14}{5} x_3 - \frac{3}{5} x_4 \end{cases}$$

$$\text{where} \quad \bar{\boldsymbol{A}} = \begin{pmatrix} -\frac{1}{5} & \frac{1}{10} \\ \frac{4}{5} & \frac{1}{10} \end{pmatrix} \quad \bar{\boldsymbol{b}} = \begin{pmatrix} \frac{3}{2} \\ \frac{7}{2} \end{pmatrix}$$

Example (4/4)

- In the dictionary above, $\mathcal{B} = \{1, 2\}$ and $\mathcal{N} = \{3, 4\}$
- The associated fractional basic feasible solution is

$$\bar{x}_1 = \frac{3}{2}, \quad \bar{x}_2 = \frac{7}{2}, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 0 \quad \text{of value} \quad \bar{z} = 17$$

- In the space of the original variables it is the vertex $(\frac{3}{2}, \frac{7}{2})$ of the feasible region of the LP relaxation

Example: Gomory fractional cuts (1/8)

- Consider again the ILP problem of Example 1 and its LP relaxation in canonical form with 2 non-negative variables and 2 linear constraints:

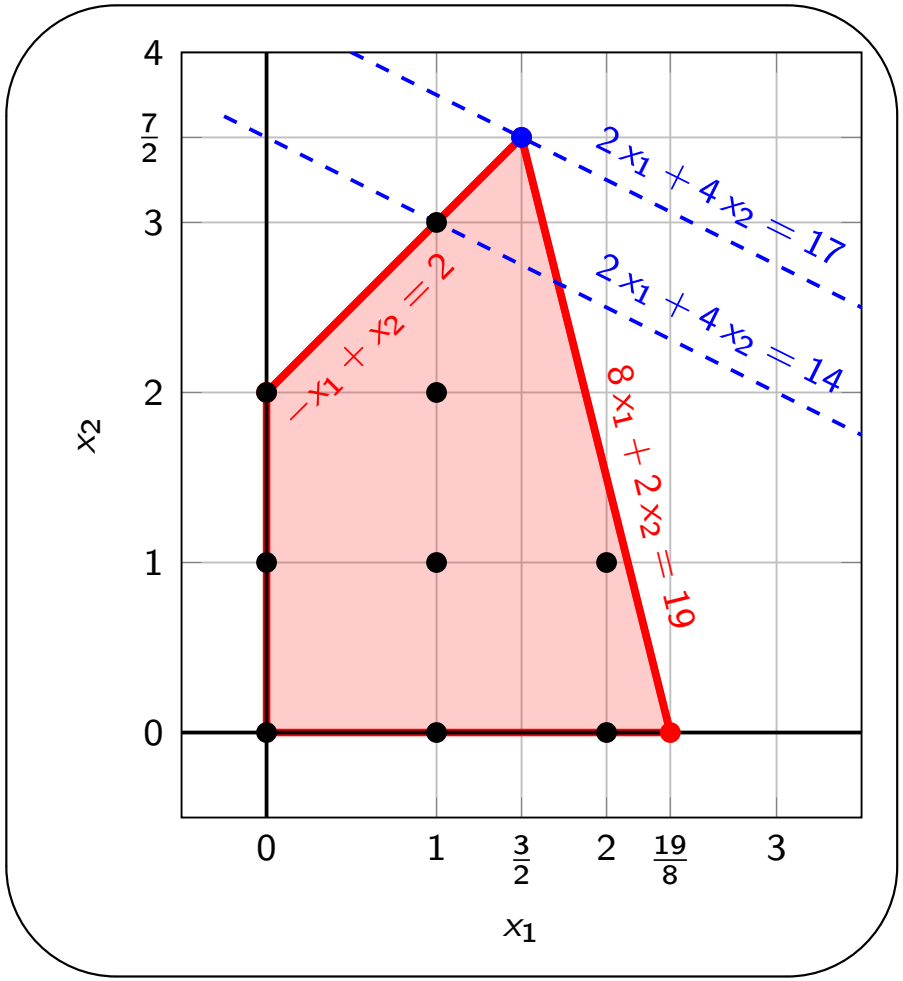
$$\begin{array}{rrcrcl} \max & 2x_1 & + & 4x_2 & & \\ & -x_1 & + & x_2 & \leq & 2 \\ & 8x_1 & + & 2x_2 & \leq & 19 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

ILP problem

$$\begin{array}{rrcrcl} \max & 2x_1 & + & 4x_2 & & \\ & -x_1 & + & x_2 & \leq & 2 \\ & 8x_1 & + & 2x_2 & \leq & 19 \\ & x_1, & & x_2 & \geq & 0 \end{array}$$

LP relaxation

Example: Gomory fractional cuts (2/8)



- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP})$ of the ILP problem is

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 3$$

with optimal value $z(\mathrm{ILP}) = 14$

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP})$ of the LP relaxation is

$$\tilde{x}_1 = \frac{3}{2}, \quad \tilde{x}_2 = \frac{7}{2}$$

with optimal value $z(\mathrm{LP}) = 17$

Example: Gomory fractional cuts (3/8)

- Consider the simplex dictionary associated with a fractional basic feasible solution of the LP relaxation in standard form of the ILP problem of Example 1:

$$\begin{cases} x_1 = \frac{3}{2} + \frac{1}{5} x_3 - \frac{1}{10} x_4 \\ x_2 = \frac{7}{2} - \frac{4}{5} x_3 - \frac{1}{10} x_4 \\ z = 17 - \frac{14}{5} x_3 - \frac{3}{5} x_4 \end{cases}$$

$$\text{where} \quad \bar{\boldsymbol{A}} = \begin{pmatrix} -\frac{1}{5} & \frac{1}{10} \\ \frac{4}{5} & \frac{1}{10} \end{pmatrix} \quad \bar{\boldsymbol{b}} = \begin{pmatrix} \frac{3}{2} \\ \frac{7}{2} \end{pmatrix}$$

- Both $\bar{b}_1$ and $\bar{b}_2$ are fractional: each of the two rows generates a Gomory fractional cut

Example: Gomory fractional cuts (4/8)

- The Gomory fractional cut generated by the first row is:

$$\underbrace{\left(-\frac{1}{5}-\left\lfloor-\frac{1}{5}\right\rfloor\right)}_{-\frac{1}{5}-(-1)=\frac{4}{5}} x_3+\underbrace{\left(\frac{1}{10}-\left\lfloor\frac{1}{10}\right\rfloor\right)}_{\frac{1}{10}-0=\frac{1}{10}} x_4 \geq \underbrace{\left(\frac{3}{2}-\left\lfloor\frac{3}{2}\right\rfloor\right)}_{\frac{3}{2}-1=\frac{1}{2}}$$

$$\frac{4}{5}\, x_3+\frac{1}{10}\, x_4 \geq \frac{1}{2} \quad \Rightarrow \quad 8\, x_3+x_4 \geq 5$$

Example: Gomory fractional cuts (5/8)

- From the standard form of the LP relaxation, we have:

$$x_3 = 2 + x_1 - x_2 \qquad \text{and} \qquad x_4 = 19 - 8\,x_1 - 2\,x_2$$

- Substituting in the Gomory fractional cut, we can express it in terms of the original variables of the ILP problem:

$$8\,(2 + x_1 - x_2) + (19 - 8\,x_1 - 2\,x_2) \geq 5 \quad \Rightarrow \quad x_2 \leq 3$$

Example: Gomory fractional cuts (6/8)

- The Gomory fractional cut generated by the second row is:

$$\underbrace{\left(\frac{4}{5} - \left\lfloor \frac{4}{5} \right\rfloor\right)}_{\frac{4}{5}-0=\frac{4}{5}} x_3 + \underbrace{\left(\frac{1}{10} - \left\lfloor \frac{1}{10} \right\rfloor\right)}_{\frac{1}{10}-0=\frac{1}{10}} x_4 \geq \underbrace{\left(\frac{7}{2} - \left\lfloor \frac{7}{2} \right\rfloor\right)}_{\frac{7}{2}-3=\frac{1}{2}}$$

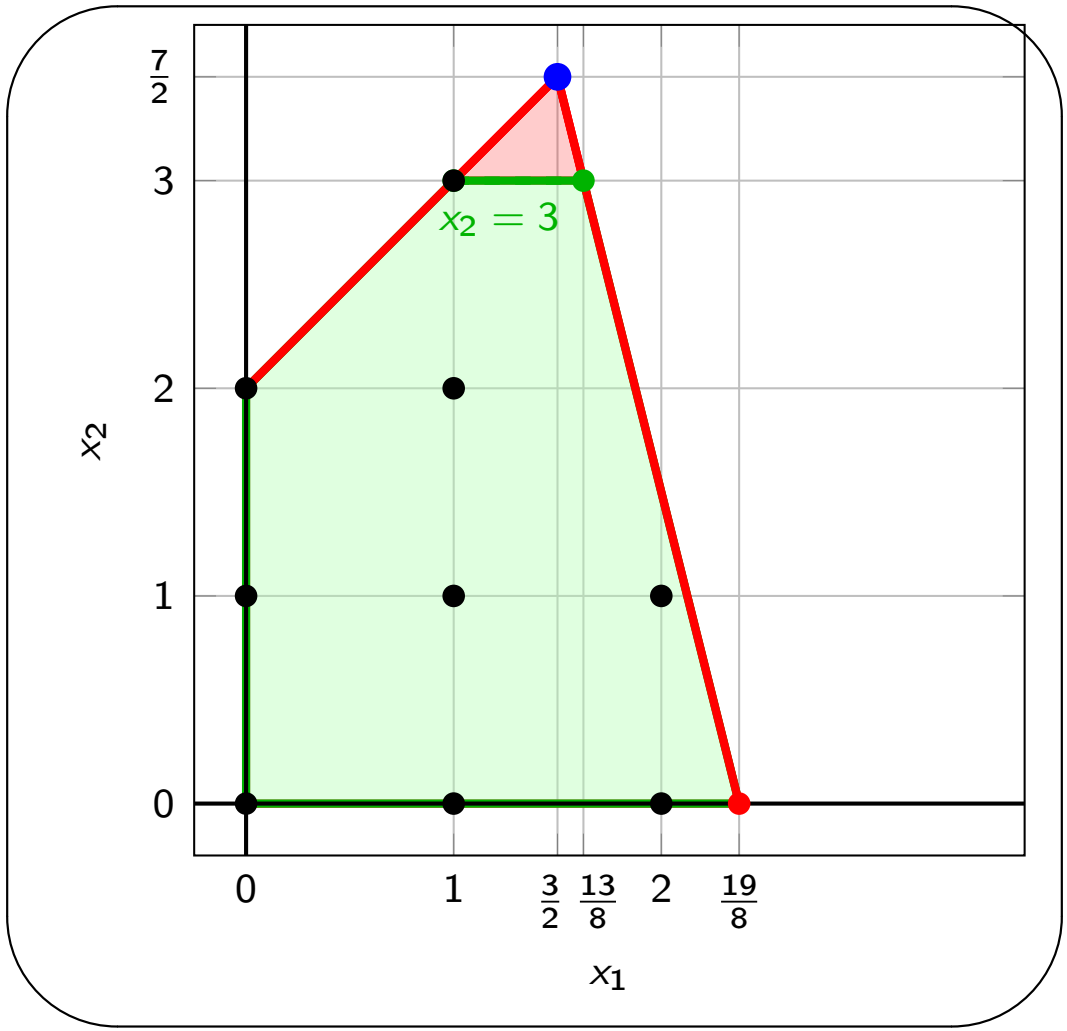
$$\frac{4}{5} x_3 + \frac{1}{10} x_4 \geq \frac{1}{2} \quad \Rightarrow \quad 8 x_3 + x_4 \geq 5$$

- This Gomory fractional cut is the same as the one generated from the first row

Example: Gomory fractional cuts (7/8)

- Effect of the Gomory fractional cut on the feasible region of the LP relaxation in canonical form (next slide)
- The cut $x_2 \leq 3$ removes the red triangle with vertices $(1, 3)$, $(\frac{3}{2}, \frac{7}{2})$ and $(\frac{13}{8}, 3)$, which contains the fractional solution $(\frac{3}{2}, \frac{7}{2})$ (blue)
- The green region that remains still contains all the integer feasible solutions (black)

Example: Gomory fractional cuts (8/8)



Effect of the cut $x_2 \leq 3$ on the feasible region of the LP relaxation in canonical form.

The objective-function row

A Gomory fractional cut can also be generated
by the **objective-function row**
of the simplex dictionary.

The cut of the objective-function row

- Consider the objective-function row of a simplex dictionary:

$$z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j x_j$$

- This row can be rewritten in the same form as the other dictionary rows:

$$z = \bar{z} - \sum_{j \in \mathcal{N}} (-\bar{c}_j) x_j$$

- The Gomory fractional cut generated by the objective-function row is

$$\sum_{j \in \mathcal{N}} \left(-\bar{c}_j - \lfloor -\bar{c}_j \rfloor\right) x_j \geq \bar{z} - \lfloor \bar{z} \rfloor$$

Example: cuts from a non-optimal dictionary (1/7)

- Consider the following simplex dictionary associated with a fractional basic feasible solution of the LP relaxation in standard form of the ILP problem of Example 1:

$$\begin{cases} x_1 = \frac{19}{8} - \frac{1}{4} x_2 - \frac{1}{8} x_4 \\ x_3 = \frac{35}{8} - \frac{5}{4} x_2 - \frac{1}{8} x_4 \\ z = \frac{19}{4} + \frac{7}{2} x_2 - \frac{1}{4} x_4 \end{cases}$$

$$\text{where} \quad \bar{\boldsymbol{A}} = \begin{pmatrix} \frac{1}{4} & \frac{1}{8} \\ \frac{5}{4} & \frac{1}{8} \end{pmatrix} \quad \bar{\boldsymbol{b}} = \begin{pmatrix} \frac{19}{8} \\ \frac{35}{8} \end{pmatrix}$$

Example: cuts from a non-optimal dictionary (2/7)

- This dictionary is **not optimal** ($\bar{c}_2 = \frac{7}{2} > 0$)
- A Gomory fractional cut can be generated from any dictionary with a fractional basic solution, optimal or not
- In the figure at the end of the example, the basic solution $(\frac{19}{8}, 0)$ is a vertex of the feasible region, not the optimum of the LP relaxation

Example: cuts from a non-optimal dictionary (3/7)

- The Gomory fractional cut generated by the first row is:

$$\underbrace{\left(\frac{1}{4} - \left\lfloor \frac{1}{4} \right\rfloor\right)}_{\frac{1}{4}-0=\frac{1}{4}} x_2 + \underbrace{\left(\frac{1}{8} - \left\lfloor \frac{1}{8} \right\rfloor\right)}_{\frac{1}{8}-0=\frac{1}{8}} x_4 \geq \underbrace{\left(\frac{19}{8} - \left\lfloor \frac{19}{8} \right\rfloor\right)}_{\frac{19}{8}-2=\frac{3}{8}}$$

$$\frac{1}{4}\, x_2 + \frac{1}{8}\, x_4 \geq \frac{3}{8} \quad \Rightarrow \quad 2\, x_2 + x_4 \geq 3$$

- From the standard form of the LP relaxation, we have $x_4 = 19 - 8\, x_1 - 2\, x_2$
- Substituting in the cut, we express it in the original variables of the ILP problem:

$$2\, x_2 + (19 - 8\, x_1 - 2\, x_2) \geq 3 \quad \Rightarrow \quad x_1 \leq 2$$

Example: cuts from a non-optimal dictionary (4/7)

- The Gomory fractional cut generated by the second row is:

$$\underbrace{\left(\frac{5}{4} - \left\lfloor \frac{5}{4} \right\rfloor\right)}_{\frac{5}{4}-1=\frac{1}{4}} x_2 + \underbrace{\left(\frac{1}{8} - \left\lfloor \frac{1}{8} \right\rfloor\right)}_{\frac{1}{8}-0=\frac{1}{8}} x_4 \geq \underbrace{\left(\frac{35}{8} - \left\lfloor \frac{35}{8} \right\rfloor\right)}_{\frac{35}{8}-4=\frac{3}{8}}$$

$$\frac{1}{4} x_2 + \frac{1}{8} x_4 \geq \frac{3}{8} \quad \Rightarrow \quad 2 x_2 + x_4 \geq 3$$

- This Gomory fractional cut is the same as the one generated from the first row

Example: cuts from a non-optimal dictionary (5/7)

- The Gomory fractional cut generated by the objective-function row is:

$$\underbrace{\left(-\frac{7}{2}-\left\lfloor-\frac{7}{2}\right\rfloor\right)}_{-\frac{7}{2}-(-4)=\frac{1}{2}} x_2 + \underbrace{\left(\frac{1}{4}-\left\lfloor\frac{1}{4}\right\rfloor\right)}_{\frac{1}{4}-0=\frac{1}{4}} x_4 \geq \underbrace{\left(\frac{19}{4}-\left\lfloor\frac{19}{4}\right\rfloor\right)}_{\frac{19}{4}-4=\frac{3}{4}}$$

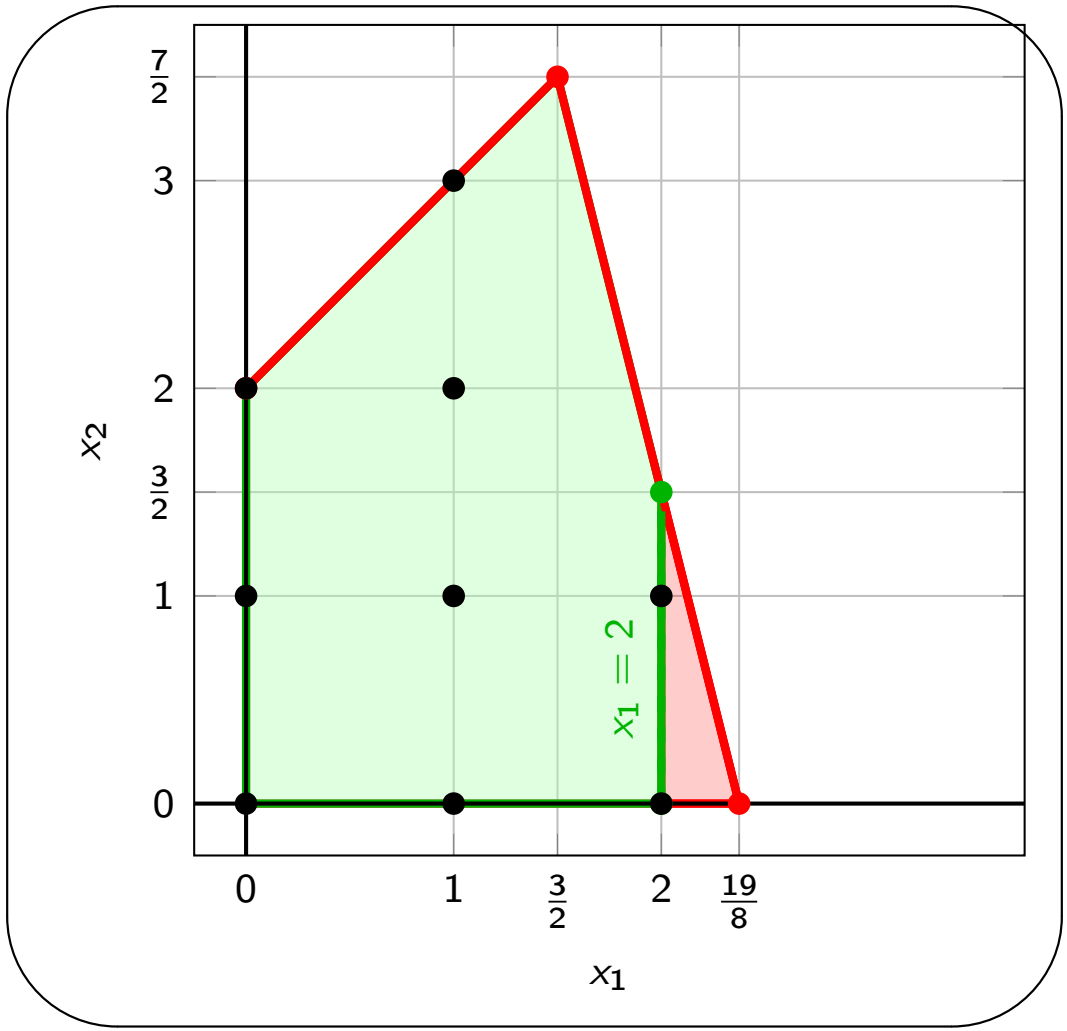
$$\frac{1}{2}\, x_2 + \frac{1}{4}\, x_4 \geq \frac{3}{4} \quad \Rightarrow \quad 2\, x_2 + x_4 \geq 3$$

- This Gomory fractional cut is the same as the one generated from the first row

Example: cuts from a non-optimal dictionary (6/7)

- Effect of the Gomory fractional cut on the feasible region of the LP relaxation in canonical form (next slide)
- The cut $x_1 \leq 2$ removes the red triangle with vertices $(2, 0)$, $(2, \frac{3}{2})$ and $(\frac{19}{8}, 0)$, which contains the basic solution $(\frac{19}{8}, 0)$
- The green region that remains still contains all the integer feasible solutions (black)

Example: cuts from a non-optimal dictionary (7/7)



Effect of the cut $x_1 \leq 2$ on the feasible region of the LP relaxation in canonical form.

Summary: Gomory fractional cuts

- A **Gomory fractional cut** is read from a row i of a simplex dictionary with fractional $\bar{b}_i$: the coefficients are the fractional parts $\bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor$, the right-hand side is $\bar{b}_i - \lfloor \bar{b}_i \rfloor$
- The objective-function row also gives a cut when $\bar{z}$ is fractional
- Different rows can give the **same** cut, and the dictionary need not be optimal
- In Example 1 the cuts are $x_2 \leq 3$ and $x_1 \leq 2$ in the original variables

We now prove the two properties of the Gomory fractional cuts:

they cut off the current fractional basic solution,
and they do not cut off any integer feasible solution.

The fractional basic solution is cut off

Observation 1

Any Gomory fractional cut cuts off the fractional basic feasible solution associated with the simplex dictionary from which it is generated.

- The idea of the proof: at the basic solution the left-hand side of the cut is zero, while its right-hand side is positive

Proof (1/2)

- In the basic feasible solution associated with the dictionary, all non-basic variables are equal to zero:

$$x_j = 0, \quad \forall j \in \mathcal{N}.$$

- Consider the row $i \in \mathcal{B}$ from which the Gomory fractional cut is generated. By definition, $\bar{b}_i$ is fractional. The corresponding Gomory fractional cut is

$$\sum_{j \in \mathcal{N}} \left(\bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor\right) x_j \geq \bar{b}_i - \left\lfloor \bar{b}_i \right\rfloor .$$

Proof (2/2)

- At the basic feasible solution, the left-hand side is equal to zero, since $x_j = 0$ for all $j \in \mathcal{N}$. On the other hand,

$$\bar{b}_i - \left\lfloor \bar{b}_i \right\rfloor > 0,$$

because $\bar{b}_i$ is fractional

- Hence, at the current basic feasible solution, the cut becomes

$$0 \not\geq \bar{b}_i - \left\lfloor \bar{b}_i \right\rfloor .$$

- Therefore, the Gomory fractional cut cuts off the basic feasible solution associated with the dictionary from which it is generated



No integer feasible solution is cut off

Observation 2

Every integer feasible solution of the ILP problem satisfies every Gomory fractional cut generated from the rows of a simplex dictionary.

- The idea of the proof: round down the coefficients of the row, then use that the left-hand side is integer to round down the right-hand side

Proof (1/4)

- Consider a Gomory fractional cut generated from the following row $i \in \mathcal{B}$ of a simplex dictionary:

$$x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij} x_j. \qquad \text{(A)}$$

- We prove that any integer feasible solution of the ILP problem satisfying this row also satisfies the corresponding Gomory fractional cut
- Since $x_j \geq 0$ for all $j \in \mathcal{N}$ and $\lfloor \bar{a}_{ij} \rfloor \leq \bar{a}_{ij}$, from (A) we obtain

$$x_i \leq \bar{b}_i - \sum_{j \in \mathcal{N}} \lfloor \bar{a}_{ij} \rfloor \, x_j.$$

Proof (2/4)

- Equivalently,
$$x_i + \sum_{j \in \mathcal{N}} \lfloor \bar{a}_{ij} \rfloor x_j \leq \bar{b}_i .$$
- The left-hand side is integer, because the solution is integer, the variables appearing in the row are integer-valued, and the coefficients $\lfloor \bar{a}_{ij} \rfloor$ are integer. Hence,
$$x_i + \sum_{j \in \mathcal{N}} \lfloor \bar{a}_{ij} \rfloor x_j \leq \left\lfloor \bar{b}_i \right\rfloor .$$

Proof (3/4)

- Using (A), we substitute

$$x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij} x_j$$

into the previous inequality and get

$$\bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij} x_j + \sum_{j \in \mathcal{N}} \lfloor \bar{a}_{ij} \rfloor x_j \leq \lfloor \bar{b}_i \rfloor .$$

Proof (4/4)

- Rearranging terms gives

$$\sum_{j \in \mathcal{N}} (\bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor) x_j \geq \bar{b}_i - \left\lfloor \bar{b}_i \right\rfloor .$$

- This is exactly the Gomory fractional cut generated from row i. Therefore, the integer feasible solution satisfies this cut
- Since both the integer feasible solution and the Gomory fractional cut were chosen arbitrarily, every integer feasible solution of the ILP problem satisfies every Gomory fractional cut generated from the rows of the dictionary

□

The objective-function row (1/3)

- The same reasoning applies to the Gomory fractional cut generated from the objective-function row. Indeed, the objective-function row can be written as

$$z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j x_j = \bar{z} - \sum_{j \in \mathcal{N}} (-\bar{c}_j) x_j.$$

- If $\bar{z}$ is fractional, the corresponding Gomory fractional cut is

$$\sum_{j \in \mathcal{N}} \left(-\bar{c}_j - \lfloor -\bar{c}_j \rfloor\right) x_j \geq \bar{z} - \lfloor \bar{z} \rfloor .$$

The objective-function row (2/3)

- This cut cuts off the current basic feasible solution: at that solution, all non-basic variables are equal to zero, and therefore the left-hand side is zero, while

$$\bar{z} - \lfloor \bar{z} \rfloor > 0.$$

- The cut is valid for every integer feasible solution of the ILP problem. Indeed, if the original objective function has integer coefficients, then z is integer-valued at every integer feasible solution. Moreover,

$$z + \sum_{j \in \mathcal{N}} \lfloor -\bar{c}_j \rfloor \, x_j \leq \bar{z}.$$

The objective-function row (3/3)

- Since the left-hand side is integer, it follows that

$$z + \sum_{j \in \mathcal{N}} \lfloor -\bar{c}_j \rfloor \, x_j \leq \lfloor \bar{z} \rfloor \, .$$

- Substituting

$$z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j x_j$$

and rearranging terms gives exactly

$$\sum_{j \in \mathcal{N}} \left(-\bar{c}_j - \lfloor -\bar{c}_j \rfloor \right) x_j \geq \bar{z} - \lfloor \bar{z} \rfloor \, .$$

Summary: validity

- At the current basic solution the left-hand side of the cut is 0 and the right-hand side is positive: the fractional solution is **cut off**
- For an integer solution, rounding down the coefficients and then the right-hand side gives exactly the cut: no integer feasible solution is cut off
- The same holds for the objective-function row, when the objective function has integer coefficients

A cut is useful only once it is part of the LP problem:
we now see how to insert it into the simplex dictionary.

Inserting a cut: three steps (1/2)

- A Gomory fractional cut is obtained from a row of the current simplex dictionary. In general, both its coefficients and its right-hand side are fractional
- To insert the cut into the simplex dictionary we proceed in three steps:
  1. substituting the definitions of the slack variables, we rewrite the cut in the original variables as an inequality $\sum_{j=1}^{n} a_j x_j \leq b$ with integer a_j and b;
  2. we make it **primitive**: we divide it by the greatest common divisor g of $a_1, \ldots, a_n$ and we round down the right-hand side,

$$\sum_{j=1}^{n} \frac{a_j}{g} x_j \leq \left\lfloor \frac{b}{g} \right\rfloor ;$$

Inserting a cut: three steps (2/2)

3. only then we add the non-negative slack variable

$$s = \lfloor b/g \rfloor - \sum_{j=1}^{n} (a_j/g)\, x_j,$$

and we substitute the current rows of the dictionary to obtain the row of s

- The rounding is valid: the left-hand side is integer for every integer $\boldsymbol{x}$
- The slack variable s is integer in every integer feasible solution of the ILP problem

First primitive, then the slack variable

First the cut is made **primitive**,
then its slack variable is added.

- A surplus variable added directly to the fractional cut corresponds to the inequality before rounding, which is weaker when g does not divide b

Example (1/2)

- Consider the ILP problem of Example 1
- The first Gomory fractional cut in the space of the original variables is the following inequality, already primitive ($g = 1$):

$$x_2 \leq 3.$$

- Adding a non-negative slack variable gives

$$x_2 + x_5 = 3, \qquad x_5 \geq 0.$$

- From the current simplex dictionary we have

$$x_2 = \frac{7}{2} - \frac{4}{5}\, x_3 - \frac{1}{10}\, x_4.$$

- Substituting and solving for the new variable x_5, the corresponding dictionary row is

$$x_5 = -\frac{1}{2} + \frac{4}{5}\, x_3 + \frac{1}{10}\, x_4.$$

Example (2/2)

- The second Gomory fractional cut in the space of the original variables is the following inequality, already primitive ($g = 1$):

$$x_1 \leq 2.$$

- Adding a non-negative slack variable gives

$$x_1 + x_5 = 2, \qquad x_5 \geq 0.$$

- From the current simplex dictionary we have

$$x_1 = \frac{19}{8} - \frac{1}{4}\, x_2 - \frac{1}{8}\, x_4.$$

- Substituting and solving for the new variable x_5, the corresponding dictionary row is

$$x_5 = -\frac{3}{8} + \frac{1}{4}\, x_2 + \frac{1}{8}\, x_4.$$

When the cut is not primitive

- In Example 1 the coefficients have no common divisor and nothing is rounded
- In the car example of the last sections the second cut in the original variables is

$$6\,x_1 + 14\,x_2 \leq 55$$

- With $g = 2$ the primitive inequality is

$$3\,x_1 + 7\,x_2 \leq \lfloor 55/2 \rfloor = 27,$$

and its slack variable is $27 - 3\,x_1 - 7\,x_2$

Summary: inclusion into the dictionary

- The cut is rewritten in the **original variables**, with integer coefficients
- It is made **primitive**: divided by the greatest common divisor g of its coefficients, with the right-hand side rounded down
- Only then its **slack variable** is added, and its row is obtained by substituting the current rows of the dictionary
- The new row has a negative value: the current basic solution is cut off

A cutting plane algorithm based on Gomory fractional cuts

Introduction: A cutting plane algorithm based on Gomory fractional cuts

- We put the cuts to work in an exact algorithm: cut, re-optimize, repeat
- We see its pseudocode, the re-optimization with the dual simplex algorithm and the rule that selects the cut
- We execute the algorithm by hand on Example 1, iteration by iteration
- We close with bounds, heuristic solutions and the limits of the method

Definition 3: cutting plane algorithm based on Gomory fractional cuts

*A **cutting plane algorithm based on Gomory fractional cuts** is an exact method for solving ILP problems based on the repeated solution of LP relaxations. Starting from the LP relaxation of the ILP problem, the algorithm progressively adds Gomory fractional cuts with the goal of removing fractional LP optimal solutions while preserving all integer feasible solutions:*

$$\rightsquigarrow \textbf{\textit{cut generation operations}}$$

After each cut generation step, the new tighter LP relaxation is solved again:

$$\rightsquigarrow \textbf{\textit{LP re-optimization operations}}$$

Two operations in a loop

- The algorithm iterates between cut generation and LP re-optimization until an optimal integer solution is found, if one exists
- We consider ILP problems in **canonical maximization form with integer input data**

Cut generation operations

- The **cut generation operations** generate a sequence of increasingly tighter LP relaxations:

$$\mathrm{LP}_1,\ \mathrm{LP}_2,\ \mathrm{LP}_3, \ldots$$

  Each LP problem is obtained from the previous one by adding one or more ***Gomory fractional cuts***

- When $k = 1$ we have the first LP problem, denoted by LP_1, which is the LP relaxation of the original ILP problem
- For $k \geq 2$, the LP_k problem is obtained from LP_{k-1} by adding the Gomory fractional cuts generated at iteration $k - 1$

Notation

For each iteration k,
if the LP_k problem has an optimal solution,
its optimal value is denoted by $z(\mathrm{LP}_k)$,
and an optimal solution is denoted by $\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$.

The upper bound

- At each iteration, the current LP relaxation contains all feasible integer solutions of the original ILP problem. Accordingly we have the following upper bound UB on the optimal value of the ILP problem:

$$z(\mathrm{ILP}) \quad \leq \quad \underbrace{\lfloor z(\mathrm{LP}_k) \rfloor}_{\mathrm{UB}}$$

- When a Gomory fractional cut is added, the feasible region of the LP relaxation becomes smaller or remains unchanged
- The algorithm may remove points that are feasible for the LP relaxation, but it does not remove any feasible integer solution of the original ILP problem
- Each new cut tightens the current LP relaxation without excluding any feasible integer solution

High-level pseudocode (1/3)

Input: An ILP problem in canonical maximization form with integer input data.

Output: An optimal solution to the ILP problem (if one exists) or a certificate of infeasibility/unboundedness.

Step 0: Initialization

Relax the integrality constraints and solve the LP_1 problem. Set the iteration counter $k \leftarrow 1$.

1. if $\tilde{\boldsymbol{x}}(\mathrm{LP}_1)$ is **integer**, then it is an optimal solution and $z(\mathrm{LP}_1)$ is the optimal value of the ILP problem. `STOP`.
2. if the LP_1 problem is **infeasible**, then the ILP problem is infeasible. `STOP`.
3. if the LP_1 problem is **unbounded**, then the ILP problem is either infeasible or unbounded. `STOP`.

High-level pseudocode (2/3)

Step 1: Cut generation

(a) Select one or more rows $i \in \mathcal{B}$ of the optimal simplex dictionary of the LP_k problem with fractional $\bar{b}_i$ and generate the associated **Gomory fractional cut(s):**

$$\sum_{j \in \mathcal{N}} \left(\bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor\right) x_j \geq \bar{b}_i - \left\lfloor \bar{b}_i \right\rfloor .$$

(b) Rewrite each cut in the original variables in primitive form, add its slack variable, and add it to the LP_k problem: this gives the new LP relaxation LP_{k+1}. Update the iteration counter $k \leftarrow k + 1$.

High-level pseudocode (3/3)

Step 2: Optimality check

(a) Solve the LP_k problem.

(b) If the LP_k problem is **infeasible**, then the ILP problem is infeasible. `STOP`.

(c) Let $\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$ be an optimal solution of the LP_k problem, and let $z(\mathrm{LP}_k)$ be its optimal value. If $\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$ is **integer**, then it is an optimal solution of the ILP problem and $z(\mathrm{LP}_k)$ is the optimal value of the ILP problem. `STOP`. Otherwise, go to Step 1.

Remarks on the pseudocode

- During the cut generation step, one can generate more than one Gomory fractional cut: by selecting more than one row of the optimal simplex dictionary with fractional $\bar{b}_i$, or the objective-function row
- The pseudocode does not say how the LP problems are re-optimized after the insertion of the cuts: the next part explains the use of the dual simplex algorithm

Re-optimization with the dual simplex algorithm

- A new cut differs from the previous LP_k problem by one constraint, so we do not solve the new problem from scratch: we start from the **optimal dictionary** of the LP_k problem and we re-optimize it with the **dual simplex algorithm**
- The cut gives a new row with the slack variable x_s of the primitive inequality in the basis
- Its value in the current basic solution is negative, since the cut removes this solution
- The reduced costs do not change

The starting point of the dual simplex algorithm

After the insertion of a cut the dictionary is **dual feasible** but not primal feasible:

this is exactly the starting point of the dual simplex algorithm.

Dual pivoting operation (1/2)

Dual pivoting operation on a dictionary

$$x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij} x_j, \quad i \in \mathcal{B}, \qquad z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j x_j,$$

with $\bar{c}_j \leq 0$ for all $j \in \mathcal{N}$:

a. the **pivot row** is the basic variable with negative value having the smallest index:

$$r = \min \left\{ \; i \in \mathcal{B} : \bar{b}_i < 0 \; \right\};$$

b. if $\bar{a}_{rj} \geq 0$ for all $j \in \mathcal{N}$, the LP problem is **infeasible**;

Dual pivoting operation (2/2)

c. otherwise the **pivot column** x_p attains the minimum ratio (ties: smallest index)

$$\frac{\bar{c}_p}{\bar{a}_{rp}} = s_m = \min\left\{ \frac{\bar{c}_j}{\bar{a}_{rj}} : j \in \mathcal{N},\ \bar{a}_{rj} < 0 \right\};$$

d. after the pivoting operation the objective value is $\bar{z} + s_m \bar{b}_r \leq \bar{z}$.

The algorithm stops when all the basic variables are non-negative: the dictionary is then optimal.

- If the new LP_{k+1} problem is infeasible, the dual simplex algorithm detects it in step b, and the ILP problem is infeasible
- Usually one or two pivoting operations are enough

We now have all the ingredients:
we execute the cutting plane algorithm by hand
on the ILP problem of Example 1.

Example: the rules of the execution

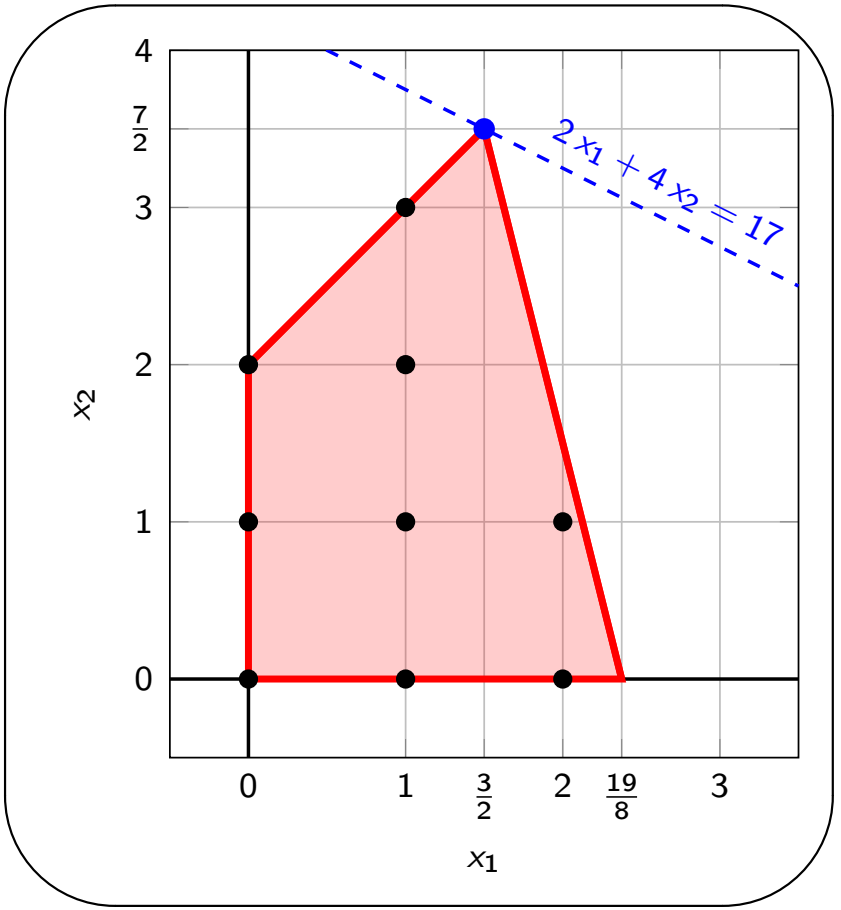
- We describe the execution of the cutting plane algorithm for the ILP problem of Example 1 using:

1. one Gomory fractional cut at each iteration;
2. the largest right-hand side as the cut selection strategy, with the tie-breaking rules of the cut selection strategies (smallest index of the basic variable, objective-function row last);
3. every LP_k problem is re-optimized with the dual simplex algorithm.

Example: initialization (1/2)

- For the LP_1 problem in canonical form, we have:

$$
\begin{array}{lrcrcl}
\max & 2\,x_1 & + & 4\,x_2 & & \\
 & -\,x_1 & + & x_2 & \leq & 2 \\
 & 8\,x_1 & + & 2\,x_2 & \leq & 19 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



Example: initialization (2/2)

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_1)$ is

$$\tilde{x}_1 = \frac{3}{2}, \quad \tilde{x}_2 = \frac{7}{2}$$

with optimal value

$$z(\mathrm{LP}_1) = 17$$

- The optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_1)$ is **fractional** and the LP_1 problem is **feasible**
- We have the upper bound on the optimal value of the ILP problem:

$$z(\mathrm{ILP}) \quad \leq \quad 17$$

Example: iteration 1, cut generation (1/6)

- We consider the following simplex dictionary associated with a fractional basic feasible solution of the LP_1 problem in standard form:

$$\begin{cases} x_1 = \frac{3}{2} + \frac{1}{5} x_3 - \frac{1}{10} x_4 \\ x_2 = \frac{7}{2} - \frac{4}{5} x_3 - \frac{1}{10} x_4 \\ z = 17 - \frac{14}{5} x_3 - \frac{3}{5} x_4 \end{cases}$$

$$\text{where} \quad \bar{\boldsymbol{A}} = \begin{pmatrix} -\frac{1}{5} & \frac{1}{10} \\ \frac{4}{5} & \frac{1}{10} \end{pmatrix} \quad \bar{\boldsymbol{b}} = \begin{pmatrix} \frac{3}{2} \\ \frac{7}{2} \end{pmatrix}$$

Example: iteration 1, cut generation (2/6)

- The Gomory fractional cut generated by the first row is:

$$\underbrace{\left(-\frac{1}{5}-\left\lfloor-\frac{1}{5}\right\rfloor\right)}_{-\frac{1}{5}-(-1)=\frac{4}{5}} x_3+\underbrace{\left(\frac{1}{10}-\left\lfloor\frac{1}{10}\right\rfloor\right)}_{\frac{1}{10}-0=\frac{1}{10}} x_4 \geq \underbrace{\left(\frac{3}{2}-\left\lfloor\frac{3}{2}\right\rfloor\right)}_{\frac{3}{2}-1=\frac{1}{2}}$$

$$\frac{4}{5}\, x_3+\frac{1}{10}\, x_4 \geq \frac{1}{2} \quad \Rightarrow \quad 8\, x_3+x_4 \geq 5$$

- From the standard form of the LP_1 problem, we have:

$$x_3 = 2 + x_1 - x_2 \qquad \text{and} \qquad x_4 = 19 - 8\, x_1 - 2\, x_2$$

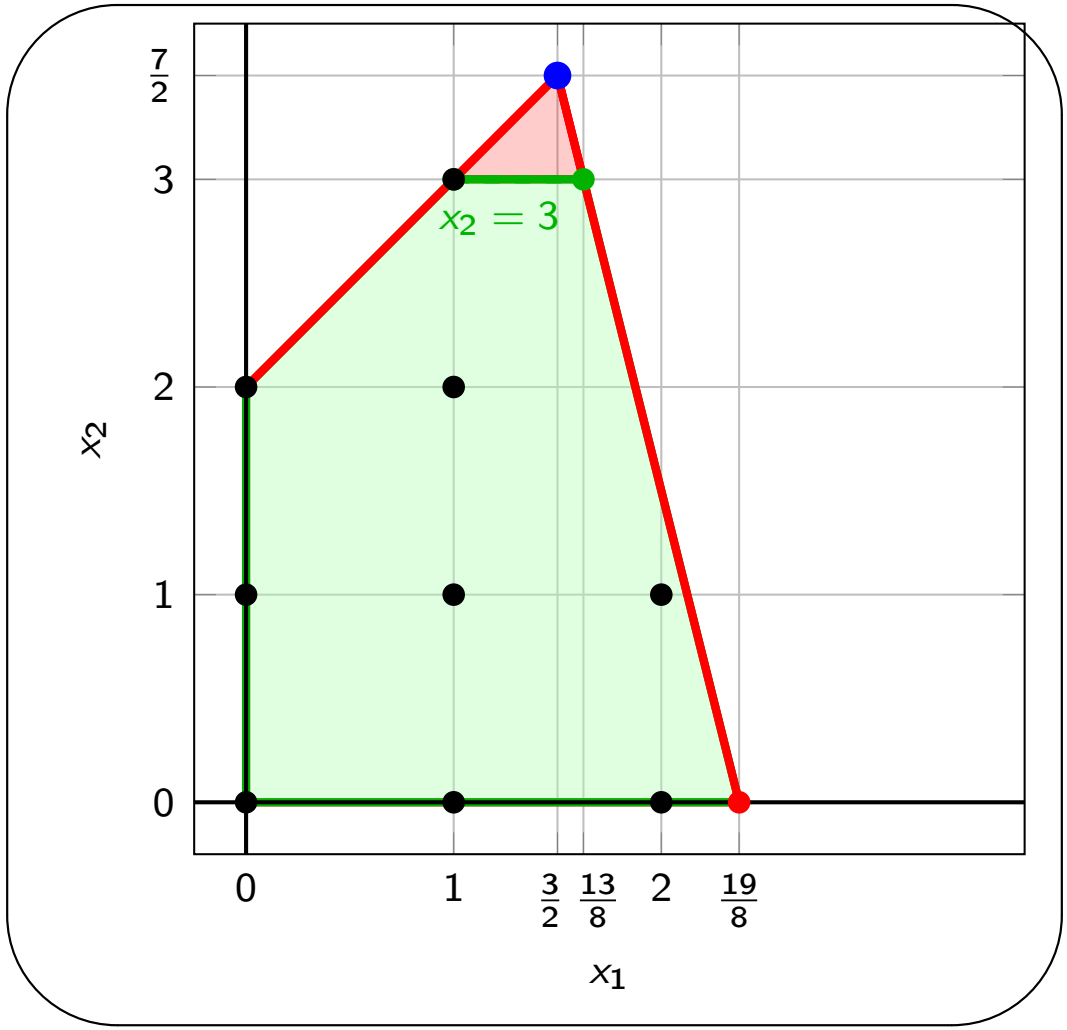
Example: iteration 1, cut generation (3/6)

- Substituting in the Gomory fractional cut, we can express it in terms of the original variables of the ILP problem:

$$8\,(2 + x_1 - x_2) + (19 - 8\,x_1 - 2\,x_2) \geq 5 \quad \Rightarrow \quad x_2 \leq 3$$

- The effect of the cut generated by the first row on the feasible region of the LP_1 problem in canonical form is shown on the next slide

Example: iteration 1, cut generation (4/6)



Effect of the cut $x_2 \leq 3$, generated by the first row, on the feasible region of the LP_1 problem.

Example: iteration 1, cut generation (5/6)

- The Gomory fractional cut generated by the second row is:

$$\underbrace{\left(\frac{4}{5} - \left\lfloor \frac{4}{5} \right\rfloor\right)}_{\frac{4}{5}-0=\frac{4}{5}} x_3 + \underbrace{\left(\frac{1}{10} - \left\lfloor \frac{1}{10} \right\rfloor\right)}_{\frac{1}{10}-0=\frac{1}{10}} x_4 \geq \underbrace{\left(\frac{7}{2} - \left\lfloor \frac{7}{2} \right\rfloor\right)}_{\frac{7}{2}-3=\frac{1}{2}}$$

$$\frac{4}{5} x_3 + \frac{1}{10} x_4 \geq \frac{1}{2} \quad \Rightarrow \quad 8 x_3 + x_4 \geq 5$$

- This Gomory fractional cut is the same as the one generated from the first row

Example: iteration 1, cut generation (6/6)

- The two cuts are equal and have the largest right-hand side, $\frac{1}{2}$
- We select the one generated by the row of x_1 (smallest index)
- We create the LP_2 problem by adding it to the LP_1 problem

Example: iteration 1, LP re-optimization (1/6)

- For the LP_2 problem in canonical form, we have:

$$
\begin{array}{lrcrcr}
\max & 2\,x_1 & + & 4\,x_2 & & \\
 & -\,x_1 & + & x_2 & \leq & 2 \\
 & 8\,x_1 & + & 2\,x_2 & \leq & 19 \\
 & & & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$

Example: iteration 1, LP re-optimization (2/6)

- In the original variables the cut is $x_2 \leq 3$, already primitive (its coefficients have no common divisor)
- The slack variable $x_5 = 3 - x_2 \geq 0$ of the primitive inequality is added
- Substituting the current rows of the dictionary, the new row is

$$x_5 = -\frac{1}{2} + \frac{4}{5}\, x_3 + \frac{1}{10}\, x_4.$$

Example: iteration 1, LP re-optimization (3/6)

- The dictionary below is **dual feasible** (the reduced costs do not change) but not primal feasible, since $\bar{b}_5 = -\frac{1}{2} < 0$:

$$\begin{cases} x_1 = \frac{3}{2} + \frac{1}{5} x_3 - \frac{1}{10} x_4 \\ x_2 = \frac{7}{2} - \frac{4}{5} x_3 - \frac{1}{10} x_4 \\ x_5 = -\frac{1}{2} + \frac{4}{5} x_3 + \frac{1}{10} x_4 \\ z = 17 - \frac{14}{5} x_3 - \frac{3}{5} x_4 \end{cases}$$

Example: iteration 1, LP re-optimization (4/6)

- **Dual pivoting operation.** The pivot row is x_5, the only basic variable with a negative value
- The ratios $\bar{c}_j / \bar{a}_{5j}$ of the non-basic variables with $\bar{a}_{5j} < 0$ are

$$\frac{\bar{c}_3}{\bar{a}_{5,3}} = \frac{-\frac{14}{5}}{-\frac{4}{5}} = \frac{7}{2}, \qquad \frac{\bar{c}_4}{\bar{a}_{5,4}} = \frac{-\frac{3}{5}}{-\frac{1}{10}} = 6.$$

- The pivot column is x_3, with $s_m = \frac{7}{2}$, and the objective value becomes

$$\bar{z} + s_m \, \bar{b}_5 = 17 + \frac{7}{2} \cdot \left(-\frac{1}{2} \right) = \frac{61}{4}$$

Example: iteration 1, LP re-optimization (5/6)

- The new dictionary is primal and dual feasible, hence optimal:

$$\begin{cases} x_1 = \frac{13}{8} - \frac{1}{8} x_4 + \frac{1}{4} x_5 \\ x_2 = 3 - x_5 \\ x_3 = \frac{5}{8} - \frac{1}{8} x_4 + \frac{5}{4} x_5 \\ z = \frac{61}{4} - \frac{1}{4} x_4 - \frac{7}{2} x_5 \end{cases}$$

Example: iteration 1, LP re-optimization (6/6)

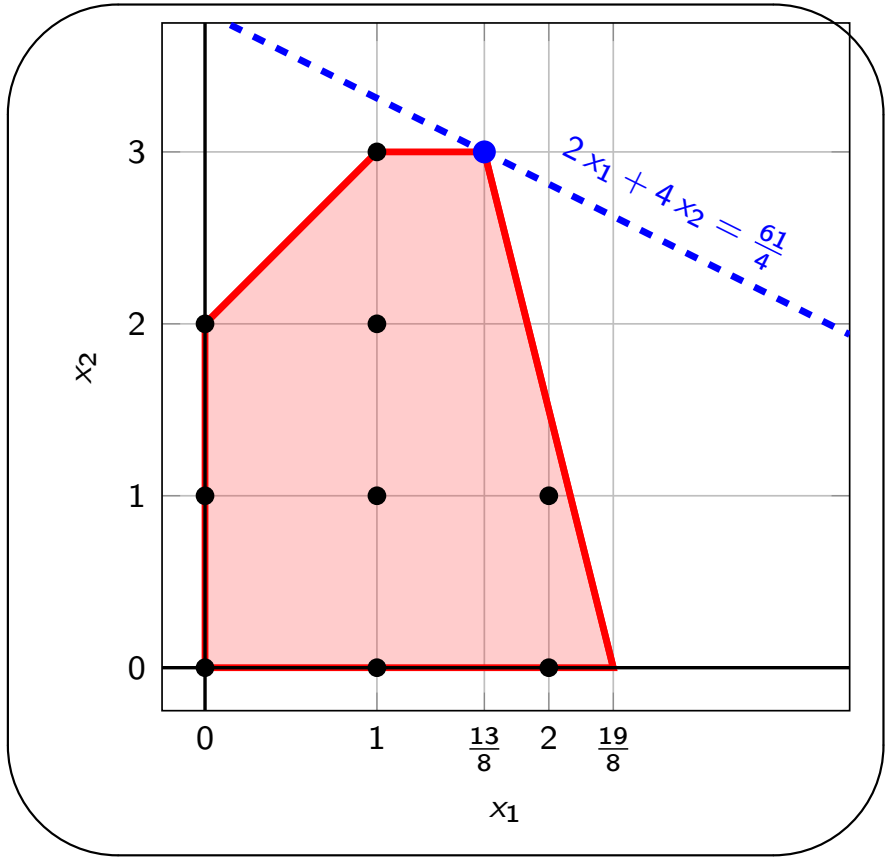
- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_2)$ is

$$\tilde{x}_1 = \frac{13}{8}, \quad \tilde{x}_2 = 3$$

with optimal value $z(\mathrm{LP}_2) = \frac{61}{4}$

- We have the upper bound on the optimal value of the ILP problem:

$$z(\mathrm{ILP}) \leq \underbrace{\left\lfloor \frac{61}{4} \right\rfloor}_{=15}$$



Example: iteration 2, cut generation (1/9)

- We have the following simplex dictionary associated with a fractional basic feasible solution of the LP_2 problem in standard form:

$$\begin{cases} x_1 = \frac{13}{8} - \frac{1}{8} x_4 + \frac{1}{4} x_5 \\ x_2 = 3 - x_5 \\ x_3 = \frac{5}{8} - \frac{1}{8} x_4 + \frac{5}{4} x_5 \\ z = \frac{61}{4} - \frac{1}{4} x_4 - \frac{7}{2} x_5 \end{cases}$$

$$\text{where} \quad \bar{\boldsymbol{A}} = \begin{pmatrix} \frac{1}{8} & -\frac{1}{4} \\ 0 & 1 \\ \frac{1}{8} & -\frac{5}{4} \end{pmatrix} \quad \bar{\boldsymbol{b}} = \begin{pmatrix} \frac{13}{8} \\ 3 \\ \frac{5}{8} \end{pmatrix}$$

Example: iteration 2, cut generation (2/9)

- The Gomory fractional cut generated by the first row is:

$$\underbrace{\left(\frac{1}{8} - \left\lfloor \frac{1}{8} \right\rfloor\right)}_{\frac{1}{8}-0=\frac{1}{8}} x_4 + \underbrace{\left(-\frac{1}{4} - \left\lfloor -\frac{1}{4} \right\rfloor\right)}_{-\frac{1}{4}-(-1)=\frac{3}{4}} x_5 \geq \underbrace{\left(\frac{13}{8} - \left\lfloor \frac{13}{8} \right\rfloor\right)}_{\frac{13}{8}-1=\frac{5}{8}}$$

$$\frac{1}{8}\, x_4 + \frac{3}{4}\, x_5 \geq \frac{5}{8} \quad \Rightarrow \quad x_4 + 6\, x_5 \geq 5$$

- From the standard form of the LP_2 problem, we have:

$$x_4 = 19 - 8x_1 - 2x_2 \qquad \text{and} \qquad x_5 = 3 - x_2.$$

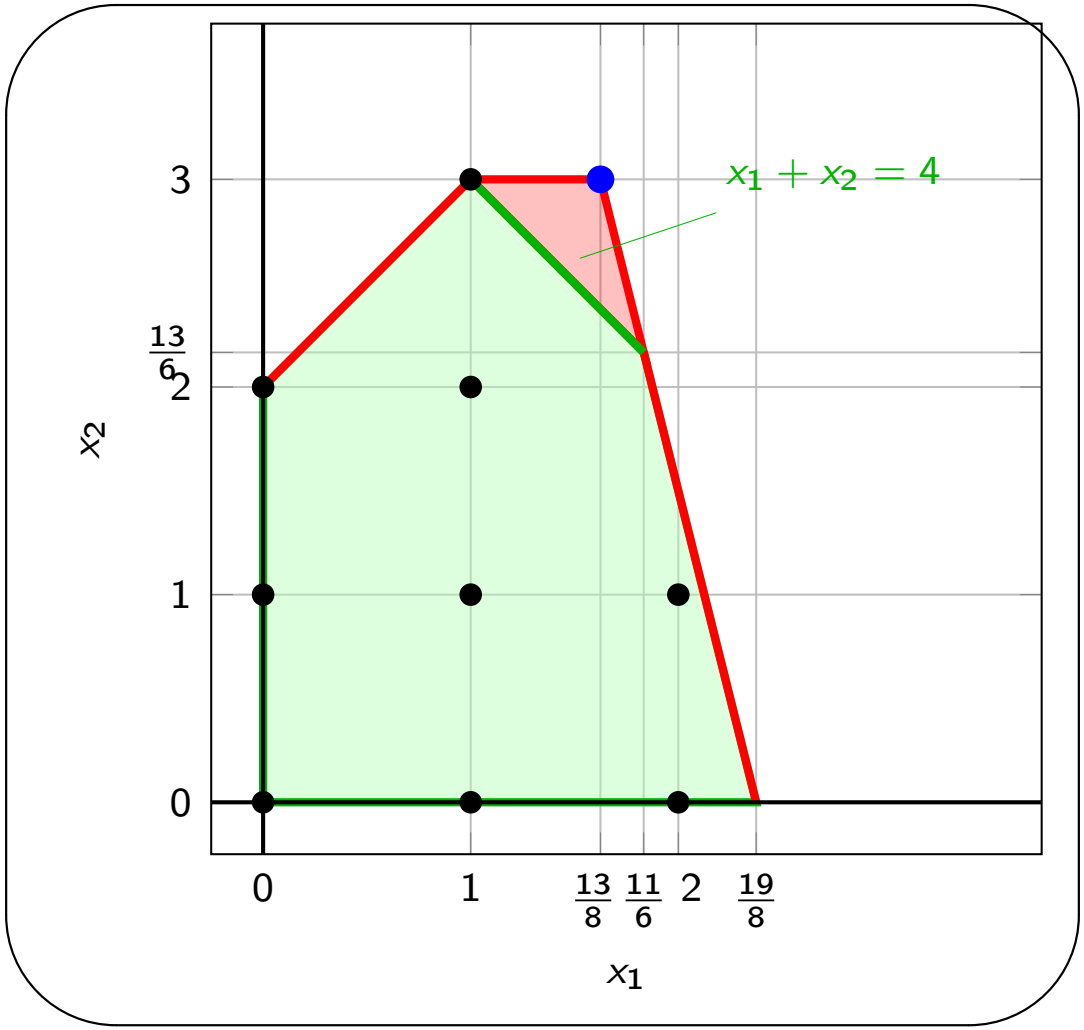
Example: iteration 2, cut generation (3/9)

- Substituting in the Gomory fractional cut, we can express it in terms of the original variables of the ILP problem:

$$(19 - 8x_1 - 2x_2) + 6(3 - x_2) \geq 5 \quad \Rightarrow \quad x_1 + x_2 \leq 4.$$

- The effect of the cut generated by the first row on the feasible region of the LP_2 problem in canonical form is shown on the next slide

Example: iteration 2, cut generation (4/9)



Effect of the cut $x_1 + x_2 \leq 4$, generated by the first row, on the feasible region of the LP_2 problem.

Example: iteration 2, cut generation (5/9)

- The Gomory fractional cut generated by the third row is:

$$\underbrace{\left(\frac{1}{8} - \left\lfloor \frac{1}{8} \right\rfloor\right)}_{\frac{1}{8}-0=\frac{1}{8}} x_4 + \underbrace{\left(-\frac{5}{4} - \left\lfloor -\frac{5}{4} \right\rfloor\right)}_{-\frac{5}{4}-(-2)=\frac{3}{4}} x_5 \geq \underbrace{\left(\frac{5}{8} - \left\lfloor \frac{5}{8} \right\rfloor\right)}_{\frac{5}{8}-0=\frac{5}{8}}$$

$$\frac{1}{8}\, x_4 + \frac{3}{4}\, x_5 \geq \frac{5}{8} \quad \Rightarrow \quad x_4 + 6\, x_5 \geq 5$$

- This Gomory fractional cut is the same as the one generated from the first row

Example: iteration 2, cut generation (6/9)

- The Gomory fractional cut generated by the objective-function row is:

$$\underbrace{\left(\frac{1}{4} - \left\lfloor \frac{1}{4} \right\rfloor\right)}_{\frac{1}{4}-0=\frac{1}{4}} x_4 + \underbrace{\left(\frac{7}{2} - \left\lfloor \frac{7}{2} \right\rfloor\right)}_{\frac{7}{2}-3=\frac{1}{2}} x_5 \geq \underbrace{\left(\frac{61}{4} - \left\lfloor \frac{61}{4} \right\rfloor\right)}_{\frac{61}{4}-15=\frac{1}{4}}$$

$$\frac{1}{4}\, x_4 + \frac{1}{2}\, x_5 \geq \frac{1}{4} \quad \Rightarrow \quad x_4 + 2\, x_5 \geq 1$$

Example: iteration 2, cut generation (7/9)

- From the standard form of the LP_2 problem, we have:

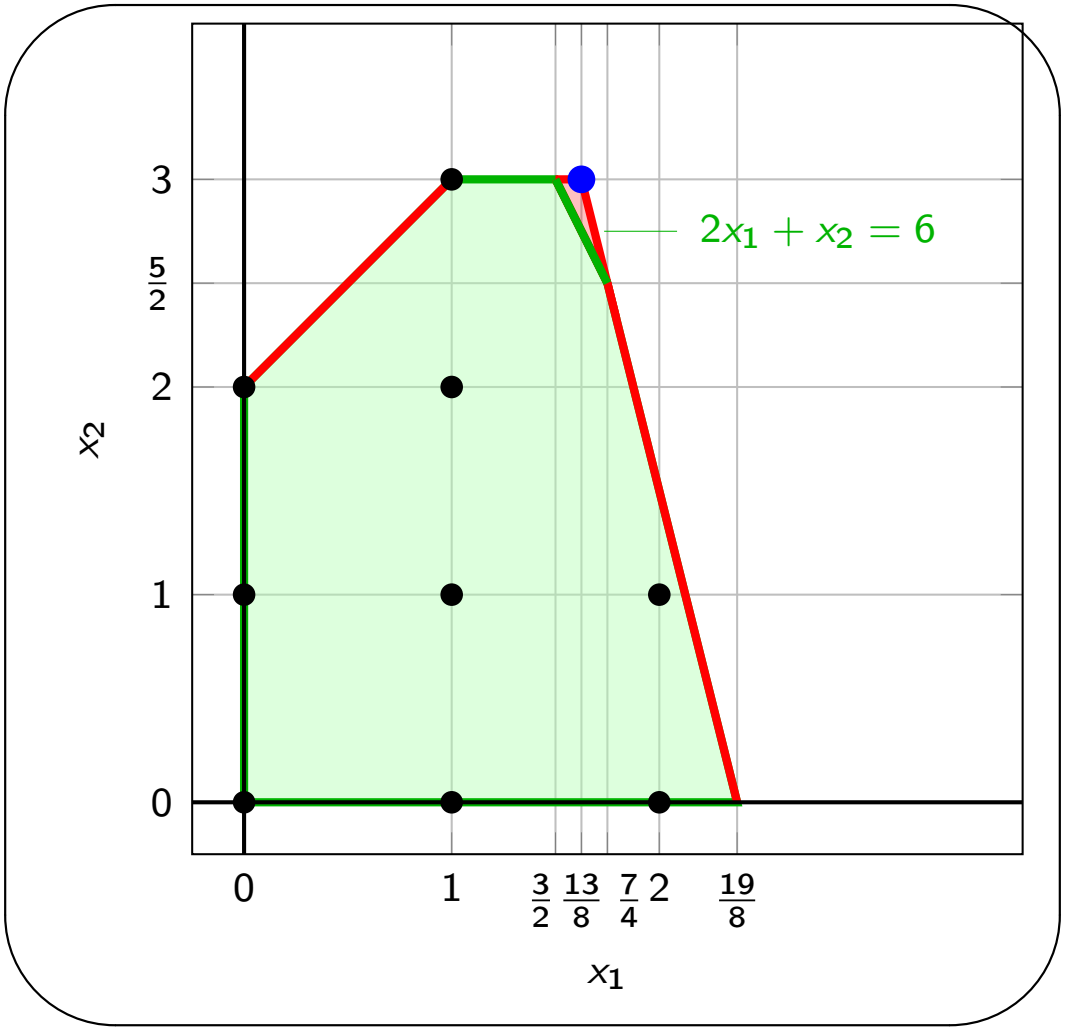
$$x_4 = 19 - 8x_1 - 2x_2 \qquad \text{and} \qquad x_5 = 3 - x_2.$$

- Substituting in the Gomory fractional cut, we can express it in terms of the original variables of the ILP problem:

$$(19 - 8x_1 - 2x_2) + 2(3 - x_2) \geq 1 \quad \Rightarrow \quad 2x_1 + x_2 \leq 6.$$

- Its effect on the feasible region of the LP_2 problem in canonical form is shown on the next slide

Example: iteration 2, cut generation (8/9)



Effect of the cut $2\,x_1 + x_2 \leq 6$, generated by the objective-function row, on the feasible region of the LP_2 problem.

Example: iteration 2, cut generation (9/9)

- The cuts of the rows of x_1 and x_3 are equal and have the largest right-hand side, $\frac{5}{8}$ (the objective-function row gives $\frac{1}{4}$)
- We select the one generated by the row of x_1 (smallest index)
- We create the LP_3 problem by adding it to the LP_2 problem

Example: iteration 2, LP re-optimization (1/6)

- For the LP_3 problem in canonical form, we have:

$$
\begin{array}{lrcrcr}
\max & 2\,x_1 & + & 4\,x_2 & & \\
 & -\,x_1 & + & x_2 & \leq & 2 \\
 & 8\,x_1 & + & 2\,x_2 & \leq & 19 \\
 & & & x_2 & \leq & 3 \\
 & x_1 & + & x_2 & \leq & 4 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$

Example: iteration 2, LP re-optimization (2/6)

- In the original variables the cut is $x_1 + x_2 \leq 4$, already primitive (its coefficients have no common divisor)
- The slack variable $x_6 = 4 - (x_1 + x_2) \geq 0$ of the primitive inequality is added
- Substituting the current rows of the dictionary, the new row is

$$x_6 = -\frac{5}{8} + \frac{1}{8}\, x_4 + \frac{3}{4}\, x_5.$$

Example: iteration 2, LP re-optimization (3/6)

- The dictionary below is **dual feasible** (the reduced costs do not change) but not primal feasible, since $\bar{b}_6 = -\frac{5}{8} < 0$:

$$
\begin{cases}
x_1 = \frac{13}{8} - \frac{1}{8} x_4 + \frac{1}{4} x_5 \\
x_2 = 3 - x_5 \\
x_3 = \frac{5}{8} - \frac{1}{8} x_4 + \frac{5}{4} x_5 \\
x_6 = -\frac{5}{8} + \frac{1}{8} x_4 + \frac{3}{4} x_5 \\
z = \frac{61}{4} - \frac{1}{4} x_4 - \frac{7}{2} x_5
\end{cases}
$$

Example: iteration 2, LP re-optimization (4/6)

- **Dual pivoting operation.** The pivot row is x_6, the only basic variable with a negative value
- The ratios $\bar{c}_j / \bar{a}_{6j}$ of the non-basic variables with $\bar{a}_{6j} < 0$ are

$$\frac{\bar{c}_4}{\bar{a}_{6,4}} = \frac{-\frac{1}{4}}{-\frac{1}{8}} = 2, \qquad \frac{\bar{c}_5}{\bar{a}_{6,5}} = \frac{-\frac{7}{2}}{-\frac{3}{4}} = \frac{14}{3}.$$

- The pivot column is x_4, with $s_m = 2$, and the objective value becomes

$$\bar{z} + s_m \, \bar{b}_6 = \frac{61}{4} + 2 \cdot \left(-\frac{5}{8} \right) = 14$$

Example: iteration 2, LP re-optimization (5/6)

- The new dictionary is primal and dual feasible, hence optimal:

$$
\begin{cases}
x_1 = 1 + x_5 - x_6 \\
x_2 = 3 - x_5 \\
x_3 = 0 + 2x_5 - x_6 \\
x_4 = 5 - 6x_5 + 8x_6 \\
z = 14 - 2x_5 - 2x_6
\end{cases}
$$

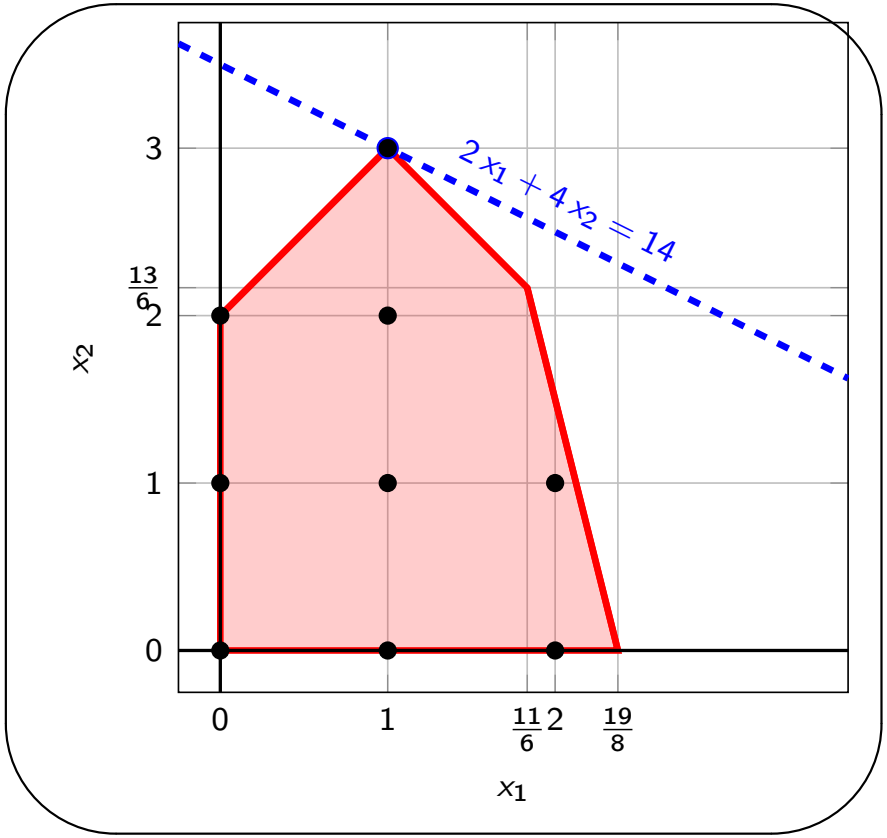
Example: iteration 2, LP re-optimization (6/6)

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_3)$ is

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 3$$

with optimal value $z(\mathrm{LP}_3) = 14$

- The optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_3)$ is **integer**



Example: the end of the algorithm

- The optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_3)$ is **integer**. Accordingly, an optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP})$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 3$$

- The optimal value $z(\mathrm{ILP})$ is 14
- Two cuts were enough: the upper bound went from 17 to 15 and then to 14

Cut selection strategies

- During the cut generation phase, several Gomory fractional cuts can be generated and added to the LP relaxation
- The **cut selection strategy** decides which ones

The **largest right-hand side cut selection strategy**
selects the Gomory fractional cut
whose fractional right-hand side is maximal
among all the candidate cuts.

The candidate cuts and the ties

- The candidate cuts are generated by the rows $i \in \mathcal{B}$ of the dictionary with fractional $\bar{b}_i$; the right-hand side of the cut of row i is
$$\bar{b}_i - \left\lfloor \bar{b}_i \right\rfloor .$$
- If the objective value $\bar{z}$ is fractional, the objective-function row is also a candidate, with right-hand side $\bar{z} - \lfloor \bar{z} \rfloor$
- **Ties.** Cuts with the same right-hand side can be different, so the choice must be fixed to make the execution reproducible: we select the row of the basic variable with the smallest index, and the objective-function row is considered last

Heuristic initialization

Before executing a cutting plane algorithm,
a **heuristic algorithm** can be run
with the goal of finding a high-quality feasible solution
to the ILP problem.

This heuristic solution provides
an initial lower bound LB on the optimal value.

Monitoring the gap

- At each iteration, the optimal value of the current LP relaxation provides an **upper bound**

$$\mathrm{UB} = \lfloor z(\mathrm{LP}_k) \rfloor$$

on the optimal value of the ILP problem. As valid cuts are added, this upper bound is progressively improved

- The availability of a feasible integer solution allows one to monitor the gap between the upper and lower bounds:

$$\frac{\mathrm{UB} - \mathrm{LB}}{|\mathrm{LB}|}$$

- The cutting plane algorithm can be stopped when this gap is zero: then the heuristic solution is proven to be optimal

Additional remarks

- The finite termination of a cutting plane algorithm with Gomory fractional cuts is guaranteed only with precise rules (a lexicographic dual simplex algorithm and a suitable choice of the rows). With arbitrary rules the algorithm may require a very large number of iterations
- Gomory fractional cuts are usually very effective in the first iterations, but their effectiveness decreases as the algorithm progresses (**tailing off**); the next section shows an example
- For this reason, Gomory fractional cuts are often used to strengthen the LP relaxations solved at the nodes of a branch-and-bound tree: the resulting method is a **branch-and-cut algorithm**

Summary (1/2)

The algorithm:

- A sequence $\mathrm{LP}_1, \mathrm{LP}_2, \ldots$ of tighter LP relaxations: cut generation and LP re-optimization in a loop
- It stops when $\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$ is integer, or when an LP_k problem is infeasible
- The upper bound is $\mathrm{UB} = \lfloor z(\mathrm{LP}_k) \rfloor$

Re-optimization:

- After a cut the dictionary is dual feasible: the **dual simplex algorithm** starts from it
- Usually one or two dual pivoting operations are enough

Summary (2/2)

Cut selection:

- Largest right-hand side; ties by the smallest index of the basic variable, objective-function row last

Bounds and limits:

- A heuristic solution gives LB, and the algorithm can stop when the gap is zero
- Finite termination needs precise rules; cuts lose effectiveness (**tailing off**) and are used inside **branch-and-cut**

Example 1:

- Two cuts, $x_2 \leq 3$ and $x_1 + x_2 \leq 4$, give the optimal solution $(1, 3)$ of value 14

The car example

Introduction: The car example

- We go back to the guiding example of the module and answer the challenge: how many cuts are needed?
- The algorithm is the same as in Example 1, with the same rules
- The car example shows the **tailing off** and the role of the primitive cuts

The rules and the table

- The car example is solved by the cutting plane algorithm with the same rules as in Example 1: one cut per iteration, largest right-hand side, ties by the smallest index of the basic variable and objective-function row last

- The table on the next slide reports, for every cut:
  - the inequality in the original variables before rounding (when $g > 1$) and in primitive form
  - the solution and the value of the next LP problem
  - the upper bound $\mathrm{UB} = \lfloor z(\mathrm{LP}_{k+1}) \rfloor$

The cuts of the car example

Cut	Before rounding	Primitive cut	$\tilde{\boldsymbol{x}}(\mathrm{LP}_{k+1})$	$z(\mathrm{LP}_{k+1})$	UB
1	—	$5\,x_1 + 14\,x_2 \leq 52$	$(\frac{211}{60}, \frac{59}{24})$	$\frac{232}{15}$	15
2	$6\,x_1 + 14\,x_2 \leq 55$	$3\,x_1 + 7\,x_2 \leq 27$	$(\frac{207}{58}, \frac{135}{58})$	$\frac{891}{58}$	15
3	—	$4\,x_1 + 7\,x_2 \leq 30$	$(\frac{65}{18}, \frac{20}{9})$	$\frac{275}{18}$	15
4	$7\,x_1 + 7\,x_2 \leq 40$	$x_1 + x_2 \leq 5$	$(\frac{25}{6}, \frac{5}{6})$	$\frac{85}{6}$	14
5	—	$9\,x_1 + 4\,x_2 \leq 40$	$(4, 1)$	14	14

The answer to the challenge

- The algorithm needs 5 cuts. The upper bound stays at 15 for the first 3 cuts, and the integer solution is found only at the last cut
- In cuts 2 and 4 the coefficients have a common divisor: the primitive cut is stronger than the inequality before rounding
- The optimal solution is $\tilde{\boldsymbol{x}} = (4, 1)$, with value $z(\mathrm{ILP}) = 14$

Summary

- On the car example the cutting plane algorithm needs 5 cuts to reach the optimal solution $(4, 1)$, of value 14
- The upper bound improves slowly: **tailing off**
- Making the cuts **primitive** matters: in cuts 2 and 4 it gives a stronger inequality

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often with Gomory fractional cuts and the cutting plane algorithm
- They concern the validity of the cut, its computation, its insertion and the execution of the algorithm
- Knowing them helps to execute the algorithm correctly in the exercises

Typical mistakes

Observation 3: Typical mistakes (1/3)

- ***Using the cut for problems with continuous variables.*** *The fractional cut is valid only if all the variables and the data are integer: with a continuous variable it may remove feasible solutions.*
- ***Taking the floor of a coefficient or of a right-hand side.*** *The cut uses the fractional parts:* $\bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor$ *is always in* $[0, 1)$*, also for a negative* $\bar{a}_{ij}$*.*

Typical mistakes

Observation: Typical mistakes (2/3)

- ***Forgetting the objective-function row.*** *It is a candidate cut if $\bar{z}$ is fractional; with the largest right-hand side strategy it can be the selected one.*
- ***Adding the surplus variable before making the cut primitive.*** *The cut in the original variables must first be divided by the greatest common divisor of its coefficients, with the right-hand side rounded down; only then is its slack variable added. Otherwise the cut is weaker.*

Typical mistakes

Observation: Typical mistakes (3/3)

- ***Choosing among equal right-hand sides without a rule.*** *Different choices give different executions; use the smallest index of the basic variable, and the objective-function row last.*
- ***Solving every*** LP_k ***problem from scratch.*** *The answer is the same, but much slower: the dictionary with the new row is dual feasible, and the dual simplex algorithm starts from it.*
- ***Stopping when the upper bound stops improving.*** *The algorithm stops only when the solution of the* LP_k *problem is integer.*

Summary

Typical mistakes:

- The cut needs integer variables and integer data
- Use the fractional parts, and do not forget the objective-function row
- First make the cut primitive, then add the slack variable
- Break ties with a fixed rule, re-optimize with the dual simplex algorithm, and stop only on an integer solution

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the car example: the bounds found so far

Summary (1/2)

- A Gomory fractional cut is generated from a row of the dictionary with fractional $\bar{b}_i$; it cuts off the current basic solution and no integer feasible solution of the ILP problem
- The cut is valid only for ILP problems with integer data, where slack and surplus variables are integer
- After each cut the LP problem is re-optimized with the dual simplex algorithm, starting from the dictionary of the previous problem

Summary (2/2)

- The upper bound is $\lfloor z(\mathrm{LP}_k) \rfloor$; the algorithm ends when the LP solution is integer
- Each cut is rewritten in the original variables, made primitive (divided by the greatest common divisor of its coefficients, right-hand side rounded down), and only then its slack variable is added
- Cuts work well at the beginning and then lose effectiveness: on the car example 5 cuts are needed

Dashboard of the car example

- The lower and upper bounds on $z(\mathrm{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Primal and dual bounds	14	15	heuristic solution $(4, 1)$, and $\lfloor z(\mathrm{LP}) \rfloor$ with $z(\mathrm{LP}) = \frac{31}{2}$
This chapter	14	14	5 Gomory fractional cuts